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# Bihar Engineering University, Patna

## B.Tech 6th Semester Exam-2023

***Code: 101602*** | ***Subject: Construction Engineering and Management*** | ***Time: 03 Hours*** | ***Full Marks: 70***

***Instructions:-***

*(i)* *The marks are indicated in the right-hand margin.*
*(ii)* *There are NINE questions in this paper.*
*(iii)* *Attempt FIVE questions in all.*
*(iv)* *Question No. 1 is compulsory.*
*(v)* *Symbols used (if any) have their usual meanings.*

***Q.1 Answer any seven question of the following:*** **[14]**

(a) An activity is

(i) The beginning or end of a specified job
(ii) An element of work entailed in the project
(iii) The movement of heavy vehicle from one place to another
(iv) The progress of work upto a certain limit

(b) The probabilistic time is given by (Where $t_o$= optimistic time, $t_p$= pessimistic time and, $t_n$= Most likely time)

(i) $(t_o + t_p + t_n)/3$
(ii) $(t_o + 2t_p + t_n)/4$
(iii) $(t_o + 4t_p + t_n)/5$
(iv) $(t_o + 4t_n + t_p)/6$

(c) Pert requires

(i) Single time estimate
(ii) Double time estimate
(iii) Triple time estimate
(iv) None of these

(d) Which of the following is a dummy activity?

(i) Excavation foundation
(ii) Laying the foundation concrete
(iii) Awaiting the arrival of concrete material
(iv) Curing the foundation concrete

(e) A Gantt chart indicates

(i) Balance of work to be done
(ii) Efficiency of project work
(iii) Comparison of actual process with the scheduled progress
(iv) Progress cost of project

(f) Negative slack occurs

(i) When latest allowable time is greater than earliest expected time
(ii) When events stick to their schedule
(iii) When deficiency of resource exist
(iv) None of these

(g) PERT calculation yield a project length of 60 week with a variance of 9 week. Probability factor corresponding to 95% is 1.647; then the number of week required to complete the project with a probability of 95% is

(i) 60.94
(ii) 62.94
(iii) 64.94
(iv) 66.9417

(h) Which of the following earth moving machines has the shortest cycle time

(i) Drag line
(ii) Hoe
(iii) Clam shell
(iv) Dipper shovel

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(i) Which of the following data is not required to prepare an estimate?

(i) Amplitude
(ii) Drawings
(iii) Specifications
(iv) Rates

(j) Force majeure can be

(i) Act of god
(ii) Fire on explosion
(iii) Wars on hostilities
(iv) All of the above

***Q.2 Solve both questions :*** **[14]**

(a) (a) Describe various phases of project management.

(b) (b) Discuss in brief the basics of slip forming for tall structure.

***Q.3 Solve this question :*** **[14]**

Following table shows details of activities of a small project: (i) Prepare AON network (ii) Prepare AOA network (iii) Identify critical path and duration of project (iv) Work out total, Free and independent floats

| Sr. No | Activity | Preceding Activity | Cost per day (Rs. Thousand) | Duration |
|---|---|---|---|---|
| 1 | A | - | 5 | 14 |
| 2 | B | - | 6 | 10 |
| 3 | C | - | 4 | 20 |
| 4 | D | A | 8 | 10 |
| 5 | E | B | 4 | 160 |
| 6 | F | C | 3 | 12 |
| 7 | G | D | 3 | 10 |
| 8 | H | D | 5 | 8 |
| 9 | I | E, F | 4 | 20 |
| 10 | J | G | 2 | 10 |
| 11 | K | H | 3 | 16 |
| 12 | L | I | 3 | 18 |

***Q.4 Solve both questions :*** **[14]**

(a) Explain the application of BIM in project management. What are the safety measures to be considered in the construction site?

(b) A target date of completion is to be forecast based on the following information: Item A take 7 weeks for completion; Item B takes 5 weeks; Item C depend on the completion of both A and B, and requires 3 weeks; Item D follow up of item C and requires 2 weeks. Prepare a Gantt chart for scheduling the completion date.

***Q.5 Solve both questions :*** **[14]**

(a) (a) Compare and contrast CPM and PERT. Under what circumstance would you use PERT as opposed to CPM in project management?

(b) (b) Discuss the objectives of network analysis.

***Q.6 Solve both questions :*** **[14]**

(a) (a) What are the factors must be taken into account before selection of construction equipment.

(b) (b) calculate the output per day of a concrete mixture of 200 litres. Assume all data reasonably for the purpose.

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***Q.7 Solve both questions :*** **[14]**

(a) (a) Discuss various rules for providing dummies in a network. What are redundant dummies?

(b) (b) What are the methods used for manufacture of precast concrete elements? Explain briefly

***Q.8 Solve both questions :*** **[14]**

(a) (a) Explain duties and responsibilities of parties involved in a contract.

(b) (b) Differentiate between conventional and mechanized methods of construction.

***Q.9 Write short notes on any four of the following:*** **[14]**

(a) Clam shell

(b) Modular construction

(c) Precast concrete construction

(d) Dewatering equipments

(e) Resource aggression and allocation

(f) Checklist for quality control

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